

Practice Test IV (Version 2)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Trigonometry).

II) Show all work.

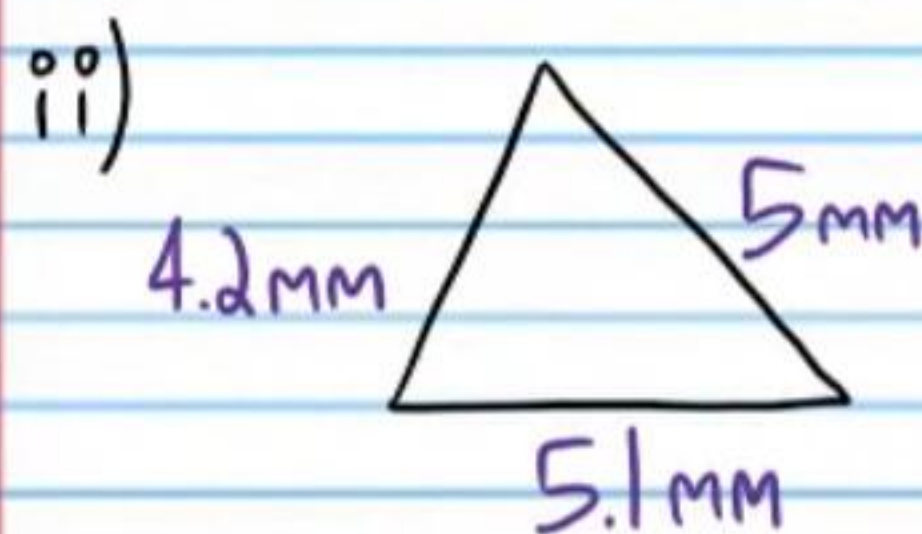
III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.

IV) You will need a calculator for this test. When π is required, use the π button on your calculator. Do not use 3.14.

V) Each problem is worth four points for a total of 100 points.

① Find the area of each triangle below.
Round to the nearest hundredth.

i) $B = 42^\circ$, $a = 9$ in, $C = 14$ in



② Find the length of an arc on a circle with radius 4 centimeters and central angle $\frac{\pi}{5}$.

③ Find the value of the following expression without a calculator.

$$\arcsin(-\sqrt{2})$$

Find the value of the expression using a calculator.
Round to the nearest thousandth.

④ $\tan\left(\frac{3\pi}{7}\right)$

⑤ Find the values of the following expressions **with a calculator**.
Round to the nearest ten thousandth.

i) $\csc 200^\circ$

ii) $\cot^{-1}(-0.9)$

⑥ i) Convert the following angle measure to radian units.

$$110^\circ$$

ii) Convert the following angle measure to degree units.

$$\frac{7\pi}{4}$$

⑦ Find the values of the following expressions **without a calculator**.

i) $\csc\left(\frac{7\pi}{6}\right)$

ii) $\tan \pi$

⑧ Solve if $0 < \alpha < 2\pi$. **Do not use a calculator**.
Write your answer in radian units.

$$\cos \alpha = -\frac{\sqrt{2}}{2}$$

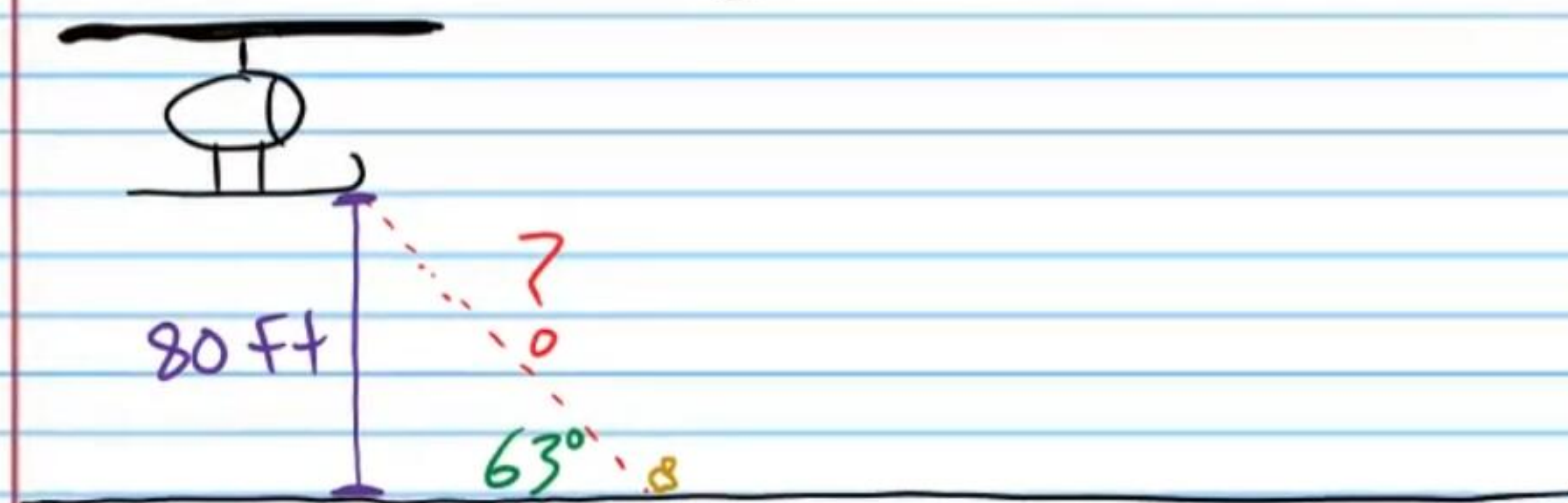
⑨ Write an equivalent expression without trigonometric notation. Assume that x is positive.

$$\sin(\sec^{-1} x)$$

⑩ Find w if $0 \leq w \leq 2\pi$. Write your answer in radian units. Do not use a calculator.

$$-5\sqrt{2} + 2\csc w = 4\csc w - 3\sqrt{2}$$

⑪ The angle of elevation from a rabbit to a helicopter is 63° . What is the shortest distance from the helicopter to the rabbit if the helicopter is 80 feet above the ground?



⑫ Solve if $0 \leq A < 360^\circ$. Round to the nearest hundredth.

$$\sec A - 3 = -\sec A$$

Find the value of each expression without a calculator.

⑬ $\cos(\sin^{-1} 1)$

$$\textcircled{14} \tan(\sec^{-1} 2)$$

$$\textcircled{15} \cot\left(\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)\right)$$

$$\textcircled{16} \sin(\sin^{-1}(-\frac{\sqrt{3}}{2}))$$

Find the value of each expression *without a calculator.*

$$\textcircled{17} \sec^{-1}(\csc 225^\circ)$$

$$(18) \text{ i) } \cos^{-1}(\cos(-\frac{\pi}{3}))$$

$$\text{ii) } \tan^{-1}(\tan \frac{\pi}{4})$$

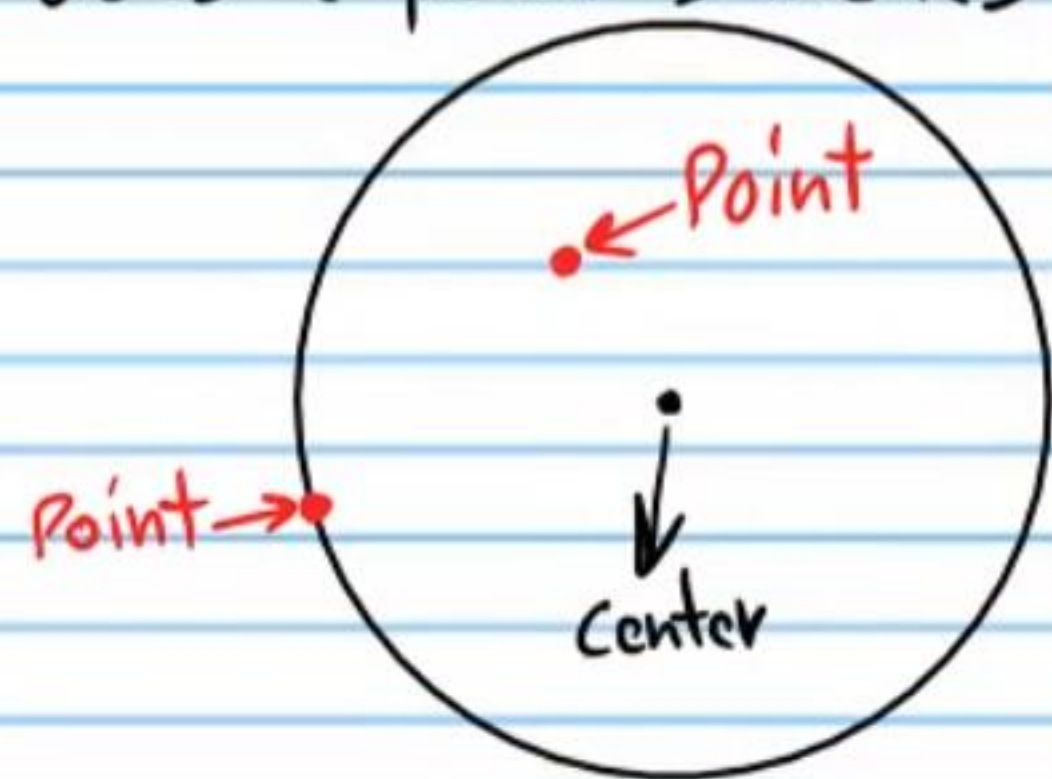
$$(19) \sin^{-1}(\cos \frac{3\pi}{2})$$

②0 A toy train moves on a circular track. It completes 10 laps in 2 minutes. If the radius of the track is 3 meters, how fast does the train move in meters per second? Round to the nearest hundredth.

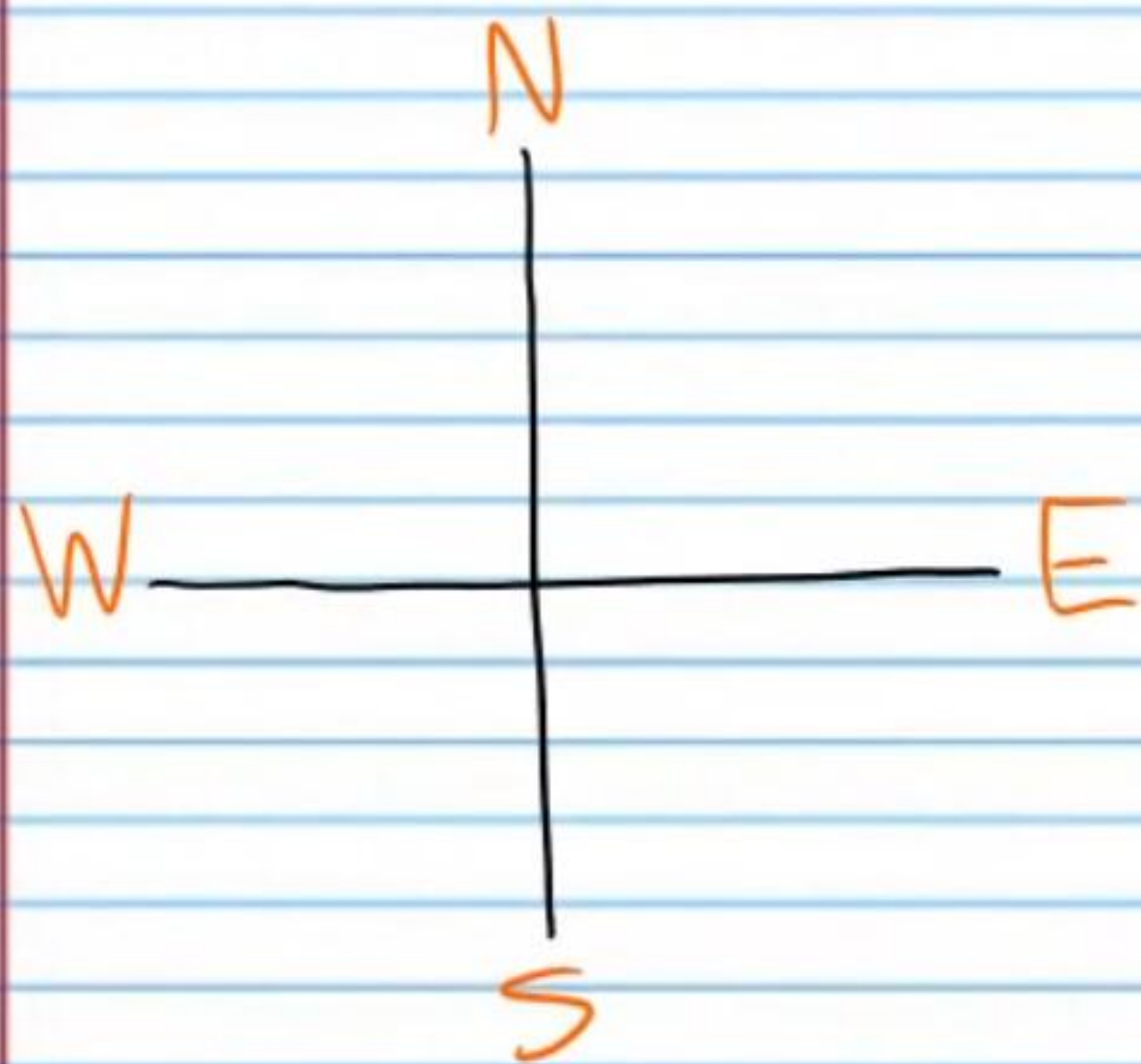
②1 A disc rotates at 94 rpm. If the diameter of the disc is 15 millimeters, how many millimeters per minute does the edge farthest from the center move? Round to the nearest whole number.

②② A wheel on a bicycle completes one full rotation in 0.08 seconds. If the diameter of the wheel is one foot, how many feet does a point on the outer edge of the wheel rotate per second? Round to the nearest thousandth.

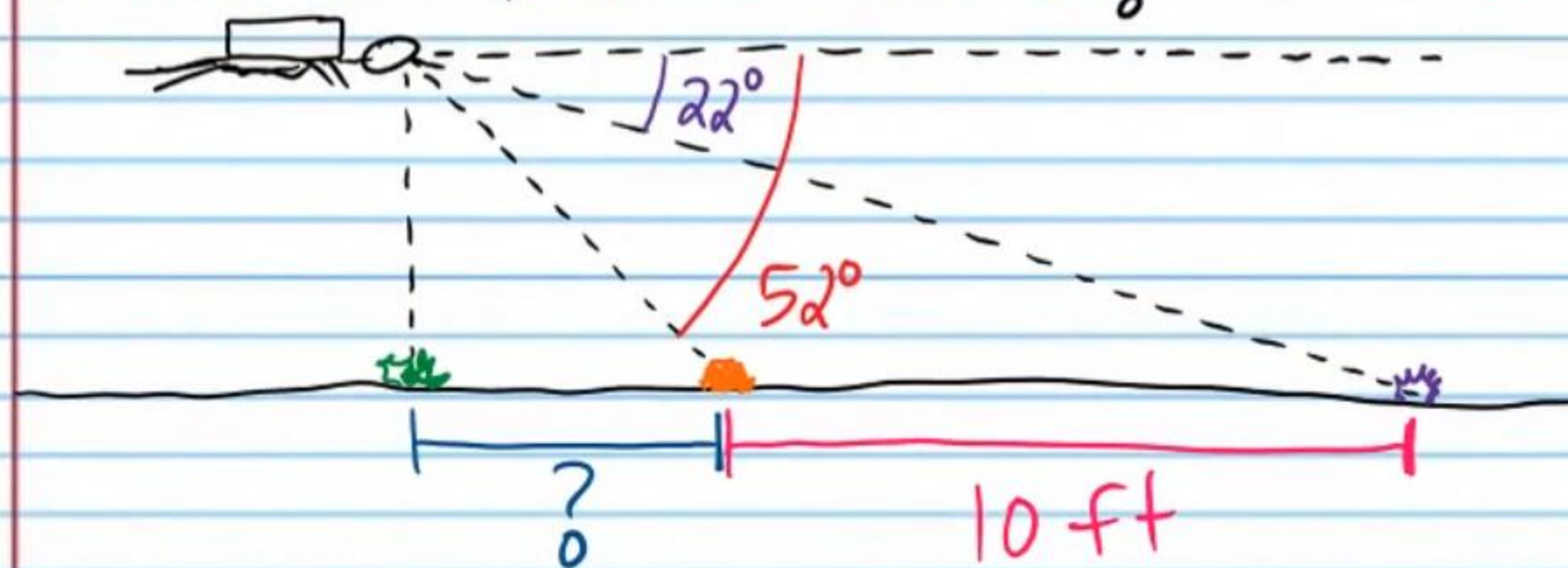
②③ A pie is placed at the center of a rotating oven. The radius of the pie is 11 inches. If a point on the outer edge moves 0.3 inches per minute, what is the angular velocity of the pie? How many inches per minute does a point 5 inches from the center move?



②4 A map shows that a person in the city of Kensington must hike 3 miles on a heading 17 degrees west of south ($S 17^\circ W$) to arrive at Blue Lake. If a person wants to hike from Kensington to the town of Lydell, he must walk 5 miles on a heading 75 degrees west of south ($S 75^\circ W$). How far apart are Blue Lake and the town of Lydell?



②5 A scuba diver looks down at a purple shell with an angle of depression of 22° . He then looks down at an orange rock with an angle of depression of 52° . If the objects are 10 feet apart, how far is the orange rock from the seaweed directly below the diver?



Practice Test IV (Version 2) Answers

① i) 42.16 in^2

ii) 9.64 mm^2

② $\frac{4\pi}{5} \text{ cm}$

③ 135°

④ 4.381

⑤ i) -1.1451

ii) 131.9872°

⑥ i) $\frac{11\pi}{18}$

ii) 315°

⑦ i) -2

ii) 0

⑧ $\frac{3\pi}{4}, \frac{5\pi}{4}$

⑨ $\frac{\sqrt{x^2-1}}{x}$

⑩ $\frac{5\pi}{4}, \frac{7\pi}{4}$

⑪ 89.8 ft

⑫ $48.19^\circ, 311.81^\circ$

⑬ 0

⑭ $\sqrt{3}$

⑮ $-\sqrt{3}$

⑯ $-\frac{\sqrt{3}}{2}$

⑰ 135°

⑱ i) $\frac{\pi}{3}$ ii) $\frac{\pi}{4}$

⑲ 0

⑳ 1.57 m/s

㉑ $4,430 \text{ mm}/\text{min}$

㉒ $39.27 \text{ ft}/\text{s}$

㉓ $\omega \approx 0.027 \text{ rad}/\text{min}, v \approx 0.136 \text{ in}/\text{min}$

㉔ 4.25 mi

㉕ 5.9 ft